

MATB34 - Written Assignment

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1 Laplace transforms

(1)

$$\mathcal{L}\{\cos^2(\omega t)\} = \mathcal{L}\left\{\frac{1}{2}(1 + \cos(2\omega t))\right\} = \frac{1}{2}(\mathcal{L}\{1\} + \mathcal{L}\{\cos(2\omega t)\}) = \frac{1}{2}\left(\frac{1}{s} + \frac{s}{s^2 + 4\omega^2}\right).$$

(2)

$$\mathcal{L}\{t^2 - 1\} = \mathcal{L}\{t^2\} - \mathcal{L}\{1\} = \frac{2}{s^3} - \frac{1}{s}.$$

(3)

$$\mathcal{L}\{e^{-t} \cos^2 t\} = \mathcal{L}\left\{e^{-t} \cdot \frac{1}{2}(1 + \cos(2t))\right\} = \frac{1}{2} \mathcal{L}\{e^{-t}\} + \frac{1}{2} \mathcal{L}\{e^{-t} \cos(2t)\} = \frac{1}{2(s+1)} + \frac{s+1}{2((s+1)^2 + 4)}.$$

2 Inverse Laplace transforms

3 Ordinary differential equations